
When Can Language Models Join Without Lexical Cues?

Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity

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<https://github.com/pankajnits/oir>

Abstract

We ask whether multi-hop answering over knowledge graphs needs readable names and predicates, or only equality among identifiers. We find that opacity blocks selection under path ambiguity, not unique-path traversal. **Opaque isomorphic reasoning (OIR)** is a measurement protocol: a renaming σ of entities and relations that we verify to be injective on each instance’s finite atom set, with no legend in the prompt. HMAC-SHA256 is one generator of σ , not a cryptographic or privacy primitive.

With the start already in the query, unique-path traversal survives sealed relation strings (OpenAI English unique 31/32, opaque unique 32/32 on a Wikidata-derived CEO slice) and both-opaque unique paths are 32/32 for Composer 2.5, OpenAI gpt-5.6-sol, and Grok 4.5. When two same-type routes compete, *relation-only* opacity (entities remain English) yields OpenAI 6/32, mostly UNKNOWN (24/32) rather than decoy (2/32). Both decoys sit on non-city P159 golds; on the 26 city-valued items the cell is 5/26 (decoy 0; McNemar 21 vs. 0). That drop is OpenAI-shaped: Grok English two-path is already 4/32. OpenAI both-opaque two-path on the same graphs is 1/32. We score the answer node, not an emitted (r_1, r_2) . Seals remove lexical cues; they are not a confidentiality proof.

1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks `works_at` then `headquartered_in` to Burbank; that success may use English predicates, or only equality among identifiers. The two-path decoy is a second person→company→city route (`partner_of/located_in`) to Kansas City. Opaque two-path keeps the English question and entity names; those predicates become HMAC tokens (`works_at` $\mapsto$ E334675310f05 on the scored Eisner item). Figure 1 is the OpenAI relation-only 2×2 (English unique 31/32; opaque two-path 6/32, mostly UNKNOWN).

OIR measures whether a model can join after English predicate strings are replaced by keyed opaque tokens. We write a 2-hop as two inferred maps (only the answer node is scored): selection $q \mapsto (r_1, r_2)$ and traversal $(v_0, r_1, r_2, G) \mapsto v_2$. Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in q as the same atom as in V (**start in q**) separates that question from start-entity identification.

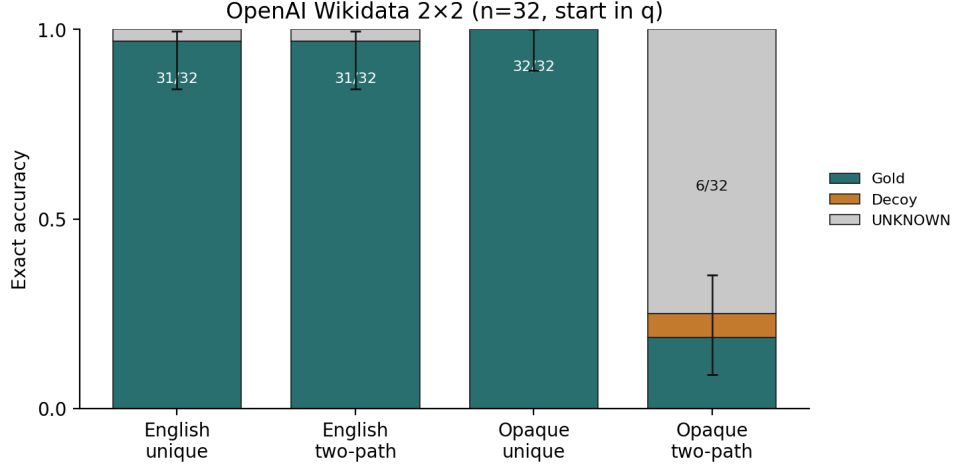


Figure 1: OpenAI WIKI-H5 on Wikidata ($n=32$, start in q ; relation-only). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals on exact match. English unique/two-path 31/32; opaque unique 32/32; opaque two-path 6/32 (mostly UNKNOWN).

Each instance uses a keyed injective σ on that instance’s atom set (HMAC-SHA256, 48-bit tokens; Section 4). The model emits the answer node in the graph’s namespace or UNKNOWN; recovering σ^{-1} is not scored. A written hop is an execution control. Matched versus novel demonstrations test in-prompt reuse of shown seals.

OIR is a measurement protocol. HMAC renaming is the same family as PrivGemo’s ϕ_Q (Tan et al., 2026); we use it as an instrument. “Lexical cues” on the primary 2×2 means relation strings (entities stay English); both-opaque is a separate ablation. English multi-hop scores should not be read as evidence that the same join succeeds once predicates are sealed and two same-type routes compete.

This paper reports: (i) unique-route sealed execution with the start in q and no written hop; (ii) end-to-end collapse when opaque relations meet path ambiguity, typically UNKNOWN rather than the decoy city; (iii) matched demonstrations restore shown seals, while novel seals do not. What is new is the interaction rather than the opacity. Privacy-oriented KGQA anonymizes entities while relations stay the readable retrieval language and reports mild degradation (Ning et al., 2026; Tan et al., 2026; Liu et al., 2026); compositional-limit and counterfactual studies vary the task and observe degradation (Dziri et al., 2023; Wu et al., 2024). Here the task, hop length, graph size, and template are fixed, only the vocabulary changes, and opacity is crossed with path ambiguity: it is free on a unique route and destructive only when two same-type routes compete, with abstention rather than a wrong route as the failure mode. Cross-dataset checks, generator robustness, and packaging probes appear in Section 5 and the appendix (Table 2).

2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next relation on a *readable* graph (Sun et al., 2024; Luo et al., 2024). Ciphred-prompt work scores in-context learning or in-cipher reasoning, not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps names in English math templates (Mirzadeh et al., 2025); OIR holds the join fixed and can hide relation strings. 2WikiMultihopQA is a readable multi-hop benchmark (Ho et al., 2020); we use English scores on our instances as the plaintext control.

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo uses HMAC ϕ_Q in a dual tower: a remote Brain ranks retrieved candidate paths and a local Hand answers on the raw graph (Tan et al., 2026). Under full anonymization they report mild degradation (CWQ 67% $\rightarrow$ 59%; WebQSP 75%). Chain-of-Relations uses relation-centric exploration to reduce dependence on incomplete entity descriptions and to avoid premature entity pruning (Liu et al., 2026); relations remain the readable retrieval language. OIR withholds retrieved paths on the free arms

Work	Hidden	Remote task	Join under opacity
ICL	substitution text	ICL task accuracy (bijection inferred)	ICL tasks
CIPHERS			
ARoG	entities; relations readable; topic names in q	retrieve + generate (relations, concepts)	privacy KG-RAG
PrivGemo	names; optional relations	rank retrieved paths; local plaintext answer	privacy KG-RAG
CoR	relations readable; entity text incomplete	relation-centric exploration	KGQA
OIR	relations (primary 2×2); entities too (ablation); one join or none	emit graph-namespaces answer (no local decode)	selection vs. traversal

Table 1: Related protocols. PrivGemo ranks retrieved paths remotely and answers locally on the raw graph. OIR sends one graph and scores the answer token from the same model.

(Table 1). The primary cell seals relations while entities stay English; both-opaque is a separate ablation. CoR and ARoG report Hits@1 on WebQSP/CWQ; we report exact match on a controlled join.

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). MaskSQL abstracts schema tokens before remote SQL (Abedini et al., 2025). Feng and Steinhardt (2024) identify binding-ID vectors that attach attributes to their entities in LM activations, and Wu et al. (2025) study binding circuitry on synthetic programs. We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound.

Compositional-limit studies change the task; we change only the vocabulary. Dziri et al. (2023) find that transformers reduce multi-step composition to linearized subgraph matching, and Wu et al. (2024) report consistent degradation once a task is moved off the default assumptions it was pretrained on. Symbol tuning replaces natural-language labels with arbitrary symbols and shows models can be pushed to rely on input-label mappings (Wei et al., 2023). Each of those interventions varies the task, the numeric base, or the label set. OIR holds the join, hop length, graph size, and question template fixed and replaces only the relation (or entity) strings, so the manipulated variable is lexical access rather than task difficulty. The 2×2 then crosses that variable with path ambiguity, which those studies do not vary: opacity is harmless on a unique route (32/32) and destructive only when two same-type routes compete. Wei et al. (2023) also motivates a control we do not run — short pronounceable symbols rather than 12-hex tokens (Section 7).

Latent-hop and shortcut work asks a related question on readable prompts. Yang et al. (2024) test whether LMs traverse two hops latently and find the second hop weaker than the first and not improving with scale; that is parametric recall over English predicates, whereas our graphs are in-context and the predicates carry no readable form, so a latent second hop has no lexical key to fire on. Multi-hop dataset design has long held that compositional questions can be answered without composing (Min et al., 2019), and MuSiQue filters single-hop shortcuts by construction (Trivedi et al., 2022). Those controls remove redundant evidence. Opacity is a complementary control: it leaves the graph and the required composition intact and removes only the lexical route from the verb in q to a predicate in G . Because unique-path opaque traversal stays at 32/32, that removal does not by itself break execution — it withdraws the cue that would otherwise select between routes.

3 Problem Formulation

Let $G = (V, R, T)$ be a labeled multi-relational graph ($T \subseteq V \times R \times V$) and q a natural-language or programmatic query with answer $a \in V$. A **seal** σ is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 is not injective over $\{0, 1\}^*$; we use 48-bit truncation because

that width is injective on our small per-instance atom sets, not as a cryptographic bound. On the scored Wikidata 2×2 instances, no HMAC collision was observed. HMAC is computed from the normalized atom string; entity and relation types are not domain-separated, so identical plaintext yields the same token regardless of type. On each scored evaluation instance the normalized entity strings and relation strings are disjoint, so no entity is identified with a relation. The model receives G or $\sigma(G)$ and a query form built from σ (or from English, depending on the arm) and must return the answer node in that namespace; recovering σ^{-1} is not scored. Sealing is deterministic *given the instance key*; the same plaintext need not map to the same token across prompts.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

In general, traversal is set-valued ($\tau: V \times R \times R \times \mathcal{G} \rightarrow 2^V$); we evaluate only instances for which $|\tau(v_0, r_1, r_2, G)| = 1$ for the gold plan (and, on two-path items, for the decoy plan as well), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in G .” We write $q \mapsto (r_1, r_2)$ when G and prompt context $\mathcal{C}$ are held fixed. A sealed join plan supplies (r_1, r_2) and tests (2). Unique-path items with v_0 already in q also test (2) when topology leaves only one route. Free sealed English tests (1) then (2) together. Dual-path items hold G fixed, give two complete routes from v_0 , and vary whether demonstrations expose the gold seals. The CEO free-English arm does *not* put v_0 in q as a single graph atom (the question is hashed piecewise), so it is not a pure test of (1).

A 2-hop is *structurally executable* when (r_1, r_2) is given, or when topology leaves only one 2-hop from a start already in q . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping supplied, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates); we do not claim mathematical unidentifiability. Without an external semantic-to-symbol map, the gold route is not explicitly identifiable from the sealed two-path graph alone; any successful selection must rely on residual statistical cues or information supplied elsewhere in the prompt. Low exact-match is therefore unsurprising when the prompt provides no explicit mapping; it is not evidence that traversal failed. Dual-path separates those cases: in-prompt reuse of demonstrated relation-seal correspondences restores end-to-end accuracy; novel seals receive no such mapping. Models may abstain (UNKNOWN) rather than guess. That is selective prediction (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017), studied in language models as calibration and abstention (Kadavath et al., 2022; Wen et al., 2025): coverage is the fraction of items with a non-UNKNOWN parse, and selective accuracy is exact match among those items, not a forced-choice among the two routes.

After sealing, the Eisner unique-path CONTEXT is Michael_Eisner | E334675310f05 | The_Walt_Disney_Company then The_Walt_Disney_Company | Eac04fab538ca | Burbank (12-hex tokens; entities remain English on the primary 2×2).

We use five measurement arms. **Readable English** is the baseline (start in q ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map; the scored output is the answer node or UNKNOWN, not (r_1, r_2) . **Sealed join plan** writes (r_1, r_2) over σ -atoms (execution control); a non-LLM router saturates on the same sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in G typically does not appear in q (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

The scored output on every free arm is that answer node (plaintext city when entities stay English; $\sigma(a)$ when entities are sealed) or UNKNOWN. We infer whether selection was supported from those end-to-end outcomes and from controls (unique topology, written hop, matched vs. novel seals); we do not score an emitted (r_1, r_2) . Throughout, “selection failure” names that inference about the latent (r_1, r_2) stage; we do not observe the pair.

4 Method

Seal recipe. Normalize each atom: strip; collapse whitespace to `_`; map other non-alphanumeric characters (except underscore, period, hyphen) to `_` using the ASCII class `A--Za--z0--9_`. (so `Zürich`→`Z_rich`; pure CJK keeps Unicode); truncate to 64; empty maps to `EMPTY`. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. Tokens are HMAC-SHA256 over UTF-8 bytes with a per-prompt 16-byte key; the token is `E` plus the first 6 digest bytes as 12 hex digits. HMAC is keyed renaming, not a security mechanism: 48-bit truncation is instance-wise injective on our small atom sets. On the scored Wikidata 2×2 graphs, normalization and HMAC produced no collisions among prompt atoms. The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess robustness to that latter variation (Table 6). The atom `Michael Eisner` normalizes to `Michael_Eisner`, so those two strings receive the same seal. Hashing the question token-by-token does not reconstruct that graph atom, which is the missing-start failure a name-binding wrapper is designed to prevent. Reported opaque cells are performance under this keyed HMAC construction, not a claim that every opaque identifier distribution yields the same rate.

Matched 2×2 (start in q). The $n=32$ suite is a seeded random sample of 32 of 73 Wikidata-derived CEO records (`random.Random(20260821).sample`; Vrandečić and Krötzsch, 2014), not a published CEO benchmark and not a hand-picked subset. Each instance is 4 triples (unique-path: 6 entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses `works_at` then the headquarters relation; the question says “led by”. The competing two-path route is synthetic (`ambig_edges` attaches record $i+1$ ’s company/HQ under `partner_of/located_in`; not a Wikidata fact; decoy HQ pool = gold HQ pool). To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length, template, and graph size fixed. Two factors vary: relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Tails are intended as the same type; Wikidata P159 was not filtered (buildings/stations/a street among gold HQs; both OpenAI opaque two-path decoys sit on city/non-city pairs; Appendix A). On Wikidata two-path graphs both intermediates are companies (`works_at/headquartered_in` vs. `partner_of/located_in`). In the primary run the gold 2-hop is listed before the decoy 2-hop in `CONTEXT`; that order is the same in the English and opaque arms. A shuffled listing of the same 32 graphs and HMAC keys (16 gold-first, 16 decoy-first) is a robustness check on a later endpoint snapshot (Appendix A): English two-path 29/32, opaque two-path 5/32 (UNKNOWN 25/32, decoy 1/32). Opaque gold-first is 3/16 vs. decoy-first 2/16: putting the decoy route first does not raise decoy selection.

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only.
No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

It does *not* contain the equality-recipe line (Appendix K). The format token is `<city_or_UNKNOWN>`, but six gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit `UNKNOWN`. The $n=32$ lock remains the primary report; Section 5.1 also scores the 26 city-valued items. English two-path is 31/32 under that `CONTEXT`-only instruction, so the line does not by itself force abstention. Readable names (Eisner, Burbank) still leave OpenAI at `UNKNOWN` 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

Further constructions. The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in q . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (Wiki-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only 2×2 . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). MOVIE-A32 and MOVIE-A100 leave English verbs such as *directed* in q ; MOVIE-B hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds G fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

Scoring. The format asks for one answer token in the graph’s namespace or UNKNOWN. Primary metric: exact match (UNKNOWN is a miss). Scored UNKNOWN is an explicit UNKNOWN line or a reply from which no answer token was parsed. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use $z=1.96$ (Wilson, 1927) ($0/32 \Rightarrow [0.00, 0.11]$; $6/32 \Rightarrow [0.09, 0.35]$; $32/32 \Rightarrow [0.89, 1.00]$). On the OpenAI Wikidata 2×2 , English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided $p=5.96 \times 10^{-8}$; McNemar, 1947). That is the two-path paired contrast, not an interaction test. That p clears Bonferroni over 60 reported cells ($\alpha/60 \approx 8 \times 10^{-4}$); we do not treat every secondary cell as a confirmatory test. **Isolation** means one quiz per file with gold answers stored outside the model-visible prompt (stricter than multi-quiz **batch** files). Isolation $n=32$ uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros. *Mid-hop error* means the model emits the sealed mid-hop company token instead of the headquarters answer.

Models. The primary roster is Composer 2.5, OpenAI gpt-5.6-sol (hereafter OpenAI), and Grok 4.5. Secondary packaging probes additionally use an unversioned Claude endpoint (Anthropic model id not stored; Appendix F). gpt-5.6-sol is the GPT-5.6 Sol Chat Completions endpoint (Sol is the flagship-equivalent tier of the GPT-5.6 family; <https://developers.openai.com/api/docs/models/gpt-5.6-sol>). OpenAI cells use that API with no tools, a single sample, and temperature omitted because that family rejects temperature 0 (max_completion_tokens 512). The runner does not pass reasoning_effort (omitted; realized effort was not logged). Composer 2.5 and Grok 4.5 cells use the same prompt files under a fixed answer format; system prompt, tools, and retry policy are not equated across families, so cross-family differences in two-path rates may partly reflect harness rather than capability. The opacity $\times$ ambiguity pattern is most interpretable for OpenAI, whose harness is the most constrained. Unique both-opaque 32/32 is the three-family traversal cell. Isolation quizzes (runs/) and locked exact-match scores (results/) are at <https://github.com/pankajjnits/oir>. Inference settings are Appendix A.

5 Results

Table 2 names the cells. WIKI-H5 is the primary discovery (OpenAI $n=32$). MOVIE-B is a cross-dataset check, not a resampling of WIKI-H5. Packaging probes are Appendix G.

ID	Dataset	Condition	n	Models	Scored
WIKI-H5	Wikidata CEO	rel-only 2×2 , start in q	32	3 APIs	city
WIKI-OO	same graphs	both-opaque	32	3 APIs	$\sigma(a)$
WIKI-SHUF	same two-path	listing order	32	OpenAI	city
MOVIE-A32	WikiMovies	actor $\rightarrow$ movie $\rightarrow$ dir vs writer	32	3 APIs	person
MOVIE-A100	WikiMovies	same hops; verbs in q	100	OpenAI	person
MOVIE-B	WikiMovies	same hops; verbs hashed	100	OpenAI	person
DUAL	WikiMovies/synth	matched vs. novel demos	32	3 APIs	$\sigma(a)$
CEO-NL	Wikidata CEO	piecewise q , no plan	32	3 APIs	$\sigma(a)$

Table 2: Cell inventory. WIKI-H5 is the discovery. A32/A100/B share the actor $\rightarrow$ movie $\rightarrow$ director vs. writer template; A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC; batch three-arm $n=12/n=24$ share one key per file.

Wikidata cell (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
English, unique	31/0/1/0	29/0/2/1	32/0/0/0
English, two paths	31/0/1/0	26/0/6/0	4/0/28/0
Opaque, unique	32/0/0/0	27/0/3/2	30/0/2/0
Opaque, two paths	6/2/24/0	10/0/22/0	1/0/31/0
Opaque two paths + plan	32/0/0/0	28/0/3/1	32/0/0/0

Table 3: Four-way outcomes: gold / decoy / UNKNOWN / other wrong token. On this table UNKNOWN is an explicit UNKNOWN line (missing/unparsed 0; both-named 0). Composer “other” cells are a wrong city that is neither gold nor decoy (English unique 1, opaque unique 2, plan 1). OpenAI’s two decoys are P159 type-asymmetric pairs. Coverage and selective accuracy for every two-path row are in the surrounding text. Grok already returns UNKNOWN on English two-path.

5.1 Opacity $\times$ path ambiguity

WIKI-H5 holds the start string in q and varies only relation lexicality and the number of 2-hops from that start. OpenAI isolation $n=32$ is the primary cell (Table 3). English unique $\approx$ English two-path $\approx$ opaque unique, while opaque two-path is much lower. A written hop on the ambiguous graph is 32/32 (control). Wilson 95%: 32/32 $\Rightarrow$ [0.89, 1.00], 31/32 $\Rightarrow$ [0.84, 0.99], 6/32 $\Rightarrow$ [0.09, 0.35]. Figure 1 plots the OpenAI 2×2 ; Table 3 gives all three families.

Let Δ_{opq} be opaque minus English exact-match. Unique path: $\Delta_{\text{opq}} = +1/32$ (32/32 vs. 31/32). Two-path: $\Delta_{\text{opq}} = -25/32$ (6/32 vs. 31/32). The descriptive difference-in-differences is $-26/32$. The primary paired comparison is two-path English versus opaque: 25 items English-correct and opaque-wrong, 0 reverse (McNemar exact $p=5.96 \times 10^{-8}$). Opacity is not a randomized within-item treatment; we report Wilson intervals and these paired counts. The primary p remains significant under Bonferroni over 60 cells; Wilson intervals treat items as independent Bernoulli trials, which the cyclic decoy construction slightly violates (Appendix A). On the 26 city-valued items (six P159 non-city HQs held out) OpenAI unique English, unique opaque, and English two-path are 26/26; opaque two-path is 5/26 (UNKNOWN 21, decoy 0; McNemar 21 vs. 0, $p=9.54 \times 10^{-7}$). Unique-path Δ_{opq} is then 0/26: the $+1/32$ on the full suite is item 3 (NRK_Marienlyst) under the city format token. Composer English two-path 26/32 is 26/26 on cities (all six misses are non-city, including Lod for gold Lod_Railway_Station). Grok English two-path is 4/26: that abstention is not a format artifact. An optional sign-flip of the per-item interaction contrast is Appendix J (same numerical p , different test). Opacity alone does not block traversal; the drop is concentrated in the two-path condition.

This pattern is *not* a universal three-family result. Grok English two-path is already 4/32 (UNKNOWN 28): ambiguity without opacity produces abstention for that family. Composer English two-path is 26/32. Opaque unique remains high (Composer 27/32, Grok 30/32); opaque two-path drops (10/32, 1/32); a written hop recovers (28/32, 32/32). Exact-match alone hides the behavior: opaque two-path is mostly UNKNOWN, not decoy. Coverage is the fraction of items with a non-UNKNOWN parse; selective accuracy is exact match among those items (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017). Two-path coverage / selective accuracy: OpenAI English 31/32, 31/31 and opaque 8/32, 6/8; Composer English 26/32, 26/26 and opaque 10/32, 10/10; Grok English 4/32, 4/4 and opaque 1/32, 1/1. A forced uniform guess among two routes would be 1/2; that is not the relevant null once UNKNOWN is allowed.

On the same graphs under two other opaque generators, unique-path accuracy exceeds two-path accuracy (permutation 31/32 vs. 15/32; alphanumeric 29/32 vs. 12/32; Table 6). Direction agrees across three generators; the two-path *rate* is not invariant.

5.2 Both-opaque and readable relations

Table 4 is a second isolation suite on the same 32 records (ANSWER_SEALED; start seal in q). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique (27/32, 30/32 in Table 3) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (Wilson [0.01, 0.16]), Composer 0/32, Grok 0/32. A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable

Arm	Ent.	Rel.	OpenAI	Composer	Grok
Unique path	opaque	Eng.	32/32	32/32	32/32
Two paths	opaque	Eng.	32/32	28/32	3/32
Unique path	opaque	opaque	32/32	32/32	32/32
Two paths	opaque	opaque	1/32	0/32	0/32
Two paths + plan	opaque	opaque	32/32	28/32	32/32

Table 4: WIKI-OO and OE on the same 32 graphs (start in q). OpenAI OE two-path 32/32; both-opaque two-path 1/32. Plan is a control.

WikiMovies people (OpenAI, $n=100$)	Unique	Two-path
English relations, Condition A (verb in q)	100/100	100/100
Opaque relations, Condition A (verb in q)	100/100	96/100
Opaque relations, Condition B (verbs hashed)	100/100	3/100
Opaque two-path + plan, Condition B	—	100/100

Table 5: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (OpenAI $n=100$, start in q). Condition B: unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; coverage 13/100, selective accuracy 3/13).

predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32.

WikiMovies unique-path with English verbs in q (MOVIE-A32) is 32/32 for all three families; opaque two-path is 17/32, 17/32, 0/32, with plan 32/32. MOVIE-A32 and MOVIE-A100 are the same actor→movie→director vs. writer template (person tails). A32’s ARM names two 2-hops; A100’s is OPAQUE_AMBIG $n=100$. On the 32 overlapping gold names OpenAI is 17/32 (15 UNKNOWN, decoy 0) vs. 29/32 under A100’s thinner ARM (all 15 UNKNOWNs become gold; three of the original golds also flip). The extra 68 items are 67/68, which is why the $n=100$ cell is 96/100. Keys differ; they cannot explain an English-name lookup. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Movie-B keeps those names, hashes the verbs, and is 3/100. The contrast is not a same-prompt Wiki-H5 replication; it remains the only header-width comparison we have (Section 7). Verbs in q can cue seals; we do not treat Condition A as a selection result.

5.3 WikiMovies: hashed verbs

Table 5 uses the same actor→movie→director vs. writer template as MOVIE-A32 (person-valued tails). Condition A leaves *directed/starred* in q (not a selection result). Condition B hashes those verbs: OpenAI unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; Wilson [0.01, 0.08]). Coverage is 13/100 and selective accuracy 3/13. A written hop is 100/100. Composer/Grok Condition A two-path: 62/100, 36/100; Condition B $n=100$ is OpenAI-only. Movie-B’s ARM names two 2-hops (Wiki-H5-style); A100 Condition A does not. The $n=32$ MetaQA-style template cell (Appendix H) is not MOVIE-B.

5.4 In-prompt seal reuse

Sealed demonstrations enable the corresponding joins when the relevant relation seals appear in the demos. Dual-path probes hold G fixed with two complete 2-hop routes (gold vs. decoy) and vary whether those gold seals appeared in the demonstrations. Matched-relation demos expose the gold relation seals; novel-relation demos teach seals that do not appear on the gold route. An explicit join plan is the execution upper bound on the same graphs. “Asymmetric” noise makes the decoy look cleaner; “balanced” noise is symmetric on both mids. Matched-relation is in-prompt reuse of demonstrated correspondences; it is not induction of a general cipher.

Isolation $n=32$ (Figure 2): OpenAI saturates matched-relation under both noise regimes (32/32, decoy 0), abstains on novel-relation balanced (UNKNOWN 32/32), and fails novel-relation asymmetric (0/32; UNKNOWN 24, decoy 8). An explicit dual-path plan scores 32/32. The same matched vs. novel contrast on WikiMovies director/writer facts is 32/32 vs. 0/32 for OpenAI, Com-

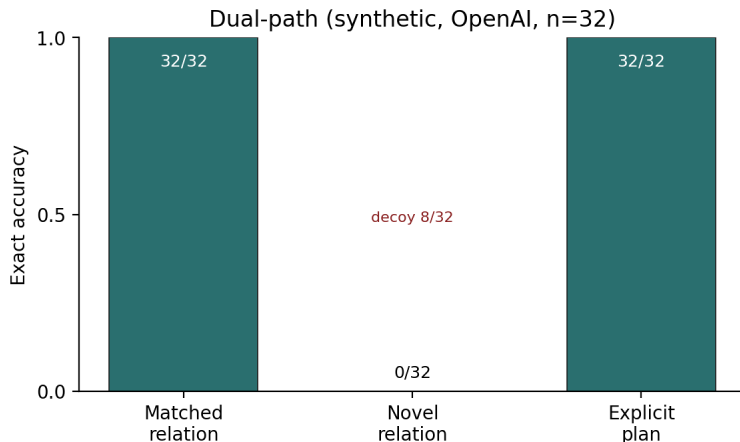


Figure 2: DUAL (synthetic, OpenAI $n=32$): matched 32/32; novel 0/32 (asymmetric decoy 8); explicit plan 32/32.

poser 2.5, and Grok 4.5 (novel is UNKNOWN except Composer decoy 2/32), with plan 32/32. Composer saturates both matched-relation arms and the plan (32/32) and, under the isolation lock (`adv_induction_n32_iso_composer25.json`, not free-form), abstains on both novel-relation arms (UNKNOWN 32/32, decoy 0). Under the free-form lock (`composer25ff`) the same novel-asymmetric items are decoy 32/32 (Table 13); the two locks differ only in answer-format strictness. Grok matches except on novel-relation asymmetric (UNKNOWN 32/32). Last-demo answer-token copy is 0/32 on every arm.

On a two-path graph a forced uniform guess among the two complete routes is $1/2$; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism. Arm definitions: Appendix D.

5.5 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5, OpenAI, and Grok 4.5 (Table 9). Batch $n=12$ and $n=24$ match that pattern. Readable English without a plan is 12/12 on CEO and 2Wiki for Composer 2.5 and OpenAI. Isolation $n=32$ is the stricter three-arm measurement: a batched free-sealed-English cell can be 0/12 while an isolated file scores if the model returns a unique neighbor without using the question text; we exclude that cell.

6 Discussion

A join is structurally executable when the route is unique or (r_1, r_2) is given (unique opaque 32/32; written hop 32/32). The prompt explicitly maps query semantics onto one relation sequence when English predicates, a legend, or demonstrated seals are present. When two complete routes exist and that mapping is absent, selection is unsupported by explicit cues: the prompt presents an underspecified interface, and the typical response is UNKNOWN rather than decoy (Table 3). We do not claim the task is mathematically unidentifiable.

For OpenAI on WIKI-H5, the difference-in-differences is $-26/32$ (25 vs. 0 paired counts). Grok already abstains on English two-path (4/32), so ambiguity alone can produce selective prediction. English relations restore OpenAI end-to-end accuracy with anonymous entities (32/32); Grok still abstains (3/32). Dual-path shows in-prompt reuse of demonstrated relation-seal correspondences and that novel seals do not transfer. The CEO piecewise-hash 0/32 is an input-binding control, not part of the matched factorial.

Factors that support execution under this construction include unique topology with the start already in q ; English predicates or an English protocol; demonstrations that reuse shown relation seals; and a same- σ equality recipe (Appendix K). A bare schema list does not (0/32; Table 11). A written join plan also saturates, as does a non-LLM router: that is a control. Hashing an English verb does not identify a sealed graph relation unless the same string is the atom in both places: `HMAC(directed)` is not `HMAC(directed_by)`.

A high English two-path rate does not imply that the same model can join once relation strings are opaque. Unique-route sealed execution can still succeed with no plan (both-opaque 32/32). Under two same-type routes and no explicit map, the typical response here is UNKNOWN, not a random decoy. When (r_1, r_2) is already known, a non-LLM router is a sufficient executor; the free sealed two-path graph is the unsupported interface. OIR diagnoses residual join ability after readable relations and retrieved paths are withheld. PrivGemo keeps those paths: full anonymization is CWQ 67% $\rightarrow$ 59% and WebQSP 75% (Tan et al., 2026). Without retrieved paths, OpenAI opaque two-path is 6/32 (coverage 8/32, selective accuracy 6/8). The two protocols are not interchangeable.

7 Limitations

WIKI-H5 is OpenAI-shaped: English two-path 31/32 drops to opaque two-path 6/32, while Grok English two-path is already 4/32. The format token `<city_or_UNKNOWN>` makes six P159 non-city golds ill-posed; the 26 city-valued items keep the OpenAI dissociation (26/26 vs. 5/26) and show that Composer English two-path 26/32 was format, not capability, while Grok 4/32 was not. Unique both-opaque 32/32 is the three-family traversal cell, not the two-path result. The suite uses a seeded sample of $n=32$ of 73 CEO records and one question template, so intervals are wide and a second seed is untested. Two-path decoys are cyclic ($i+1 \bmod 32$), so items are weakly dependent; Wilson intervals that assume independent Bernoulli trials are therefore slightly optimistic, while the within-item McNemar contrast is unaffected. English entities leave parametric cues; WIKI-OO is a complement, not a synthetic-ID experiment. HMAC two-path rates are not invariant across generators (6/32 vs. 15/32 vs. 12/32). That spread means the opaque two-path rate is a property of this token family and not of opacity alone: a 12-hex seal also costs more tokens and is harder to copy than a readable predicate. The control that would separate lexical access from token trackability — short pronounceable symbols, in the spirit of symbol tuning (Wei et al., 2023) — is not run here. Selection (r_1, r_2) is inferred from the answer node; we do not score an emitted relation pair. Dual-path tests in-prompt reuse of shown seals, not a binding mechanism. Composer 2.5 and Grok 4.5 share prompt files but not a harness, so cross-family two-path gaps may partly reflect tools and retry policy. Evidence is behavioral: we do not claim a circuit-level account of abstention. Selective accuracy stays high in every two-path opaque cell (6/8, 10/10, 1/1), so the measured effect is concentrated in coverage rather than in conditional correctness. That is an abstention profile (Kadavath et al., 2022; Wen et al., 2025), and it does not license the stronger claim that the join became impossible; a model that commits on a sealed two-path graph is usually right, which is itself evidence of residual cues we have not enumerated. Seals remove lexical cues on the instance; they are not a confidentiality proof.

The two-path ARM header names topology and opacity. English two-path 31/32 shows the header alone does not force abstention, not that header $\times$ opacity is zero. MOVIE-A32 vs. MOVIE-A100 is a confounded header-width contrast: on the 32 overlapping gold names OpenAI is 17/32 vs. 29/32 (all 15 UNKNOWNs become gold; three original golds flip); 96/100 is the $n=100$ cell because the extra 68 items are 67/68. A header-neutral Wiki-H5 rerun is the highest-value missing cell. A world-knowledge-allowed prompt, forced-choice, scored (r_1, r_2) , and an HQ-city swap are also not in this evaluation. MOVIE-B is OpenAI-only.

Ethical considerations

Instances use public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the MetaQA/WikiMovies movie graph as publicly released with WikiMovies (Miller et al., 2016) and MetaQA (Zhang et al., 2018). Appendix cells additionally use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020) and Spider (CC-BY-SA-4.0; Yu et al., 2018). No personally sensitive information is used. Model generations are not released as a corpus. Isolation quizzes (runs/) and exact-match scores (results/) are at <https://github.com/pankajmits/oir>.

8 Conclusion

Unique-route sealed execution succeeds with the start in q (opaque unique OpenAI 32/32; both-opaque unique 32/32 on three APIs). When two same-type routes compete, relation-only opacity is OpenAI 6/32 (mostly UNKNOWN); both-opaque two-path is 1/32. That is an underspecified interface, not a scored (r_1, r_2) and not a proof that traversal failed. English predicates, an equality recipe, or matched demonstrations restore end-to-end accuracy.

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A Protocol, scoring, and models

Isolation quizzes (runs/) and locked scores (results/) are at <https://github.com/pankajnits/oir> (Python ≥ 3.10). OpenAI cells: gpt-5.6-sol, no tools, one sample, August 2026 snapshot, temperature omitted, max_completion_tokens 512; reasoning_effort was omitted, and the runner stores the parsed answer, not the API model string or usage (including reasoning-token counts). WIKI-SHUF is a 7 September 2026 snapshot on the same 32 keys. Composer 2.5 and Grok 4.5 use the same prompt files; settings are not equated. WIKI-H5 uses seed 20260821 and a per-item 16-byte key derived from oir-fact-2x2-iso-v1 and the item index. The 32 records are random.Random(20260821).sample from the 73-row CEO file; two-path decoys attach record $i+1 \bmod 32$, so each item’s decoy company/HQ is the next item’s gold. Wilson intervals in the body treat items as i.i.d.; that is slightly optimistic under this cycle. A SHA-256 lockfile over every results/*.json is results/SHA256SUMS; verify with shasum -c results/SHA256SUMS. WIKI-OO is a second suite on those records (oir-entrel-2x2-iso-v1 and index). MOVIE-A32/MOVIE-A100/MOVIE-B and Dual isolation items are per-item keys; Dual demonstrations reuse that item’s key. CEO-NL isolation $n=32$ uses seed 20260820 and per-item keys. The $n=12$ three-arm is **batch** with one shared key (oir-ceiling-three-arm-v1, seed 20260728) for Composer 2.5, OpenAI, and Grok 4.5; $n=24$ is the same protocol at a larger n . We never impute n.r. as zeros.

P159 gold HQs that are not cities: Broadway, NRK_Marienlyst, Fox_Plaza, CNN_Center, Lod_Railway_Station; New_Jersey is a state. OpenAI opaque two-path decoys are items 27 and 31 (CNN_Center vs. Swindon; Lod_Railway_Station vs. Burbank). Both decoys are on the non-city subset; city-valued OpenAI opaque two-path is gold 5/26, decoy 0, UNKNOWN 21. Every non-two-path miss in Table 3 for all three families falls on those six golds. Source slice: Kansas_City 10/73, Burbank 6/73; Disney three times in the 32. An archived Movie-B ARM banner naming director vs. writer is OpenAI two-path 25/100; the locked topology-only Condition B header is 3/100.

The CEO three-arm graphs add extra edges on the gold company (owned-by, partner-of, meta-of). One question template is used. Free English hashes the question piecewise, so the full person atom typically does not appear in q .

B Generator robustness

Opaque generator (OpenAI, start in q)	Unique	Two-path
Keyed HMAC (E+12hex)	32/32	6/32
Random permutation (E+12hex)	31/32	15/32
Random alphanumeric (not HMAC-shaped)	29/32	12/32

Table 6: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson: 29/32 $\Rightarrow$ [0.76, 0.97], 12/32 $\Rightarrow$ [0.23, 0.55], 15/32 $\Rightarrow$ [0.31, 0.64]. Alphanumeric two-path is all abstention or gold (decoy 0).

A random permutation of relation labels (HMAC-shaped tokens, not keyed HMAC) on the same Wikidata 2 $\times$ 2 preserves unique-path accuracy exceeding two-path accuracy: OpenAI 31/32 vs. 15/32 (plan 30/32); Composer 28/32 vs. 2/32 (plan 28/32); Grok 31/32 vs. 0/32 (plan 32/32).

An English-entity ambiguity sweep $k=1..5$ is family-dependent: OpenAI stays high (29/32, 29/32, 28/32, 27/32, 28/32; plan 32/32); Composer declines (27/32 to 11/32; plan 29/32); Grok is 29/32 at $k=1$ then 0/32 for $k\geq 2$ (plan 32/32). Names such as Disney remain, so k is not isolated from entity semantics; the OpenAI-flat curve is not a replication of both-opaque 1/32.

C Hypothesis status and maps

Arm	Input to the model	Maps tested	Score
English, unique path	readable q , start in q	(1)+(2) (e2e)	31/32
English, two paths	readable q , start in q	(1)+(2) (e2e)	31/32
Opaque, unique path	start in q ; one route; no plan	(2) (e2e)	32/32
Opaque, two paths	start in q ; two routes; no explicit map	e2e; (1) unobserved	6/32
Plan (control)	opaque q plus (r_1, r_2)	(2)	32/32
Binding fail	piecewise q ; start typically absent	start ID (confounded)	0/32

Table 7: WIKI-H5 maps (OpenAI unless noted). Exact match on the answer node, not (r_1, r_2) . Last row is CEO-NL.

Hypothesis	Test	Status
H1. Opacity alone blocks traversal	unique-path, start in q	not supported under this construction (32/32)
H2. Written hop restores e2e (control)	two-path + plan; SealRouter	control (32/32)
H3. Input-binding control (CEO piecewise q)	CEO isolation NL	control (0/32); not selection
H4. Matched demos restore shown seals	dual-path matched vs. novel	supported (32/32 vs. 0/32)
H5. End-to-end accuracy degrades when opaque relations meet path ambiguity	matched 2×2 , start in q	supported for OpenAI (25 vs. 0)
H6. Movie KB, hashed verbs (check)	WikiMovies Cond. B $n=100$	OpenAI two-path 3/100
H7. English relations can suffice with opaque entities	Wikidata OE vs. OO	supported for OpenAI under this construction

Table 8: Hypothesis status. H5 is WIKI-H5 (OpenAI). H6 is MOVIE-B. H7 is OpenAI-specific (OE vs. WIKI-OO).

CEO-NL isolation $n=32$	Composer 2.5	OpenAI	Grok 4.5
Sealed plan / SealRouter	32/32	32/32	32/32
Free sealed English (piecewise q)	0/32	0/32	0/32

Table 9: Missing-start control, not WIKI-H5. Batch $n=12/n=24$ and packaging rows: Table 13.

Cell (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
MOVIE-A32 opaque unique	32/32	32/32	32/32
MOVIE-A32 opaque two-path	17/32	17/32	0/32
MOVIE-A32 two-path + plan	32/32	32/32	32/32
Both-opaque unique	32/32	32/32	32/32
Both-opaque two-path	1/32	0/32	0/32
Both-opaque two-path + plan	32/32	28/32	32/32

Table 10: Three-family isolation beyond Table 3. MOVIE-A32 leaves verbs in q (not a selection result). Both-opaque is Table 4 compactly.

D Dual-path arms

We name the arms by what the demos teach. **Matched-relation** (asymmetric or balanced noise): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced noise): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan**: gold relation sequence is written in the prompt.

An $n=8$ cell is zero demos 2/8; one or more demos 8/8. Isolation $n=32$ is the reported dual-path measurement. Local `mistral/qwen35` runs are not in the roster (matched-asymmetric 0/32; Qwen matched-balanced 7/32; plan n.r.).

E No-hop binding (JSON)

Arm (no written hop)	OpenAI	Composer 2.5	Grok 4.5
Schema list only	0/32	0/32	0/32
English protocol (no hop)	32/32	32/32	32/32
Same σ , silent	6/32	19/32	31/32
Same σ + equality recipe	32/32	32/32	32/32
JSONKEY : match broken	0/32	0/32	0/32
Sealed Q, two routes, recipe	31/32	25/32	32/32
Sealed Q, two routes, vague hints	13/32	23/32*	31/32

Table 11: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- σ overlap and vague hints are family-dependent, not a three-family restore. *Composer vague-hint distractor 1/32. JOIN was not evaluated at $n=32$. Wilson: 32/32 $\Rightarrow$ [0.89, 1.00], 0/32 $\Rightarrow$ [0.00, 0.11], 6/32 $\Rightarrow$ [0.09, 0.35], 31/32 $\Rightarrow$ [0.84, 0.99]. Silent same- σ OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

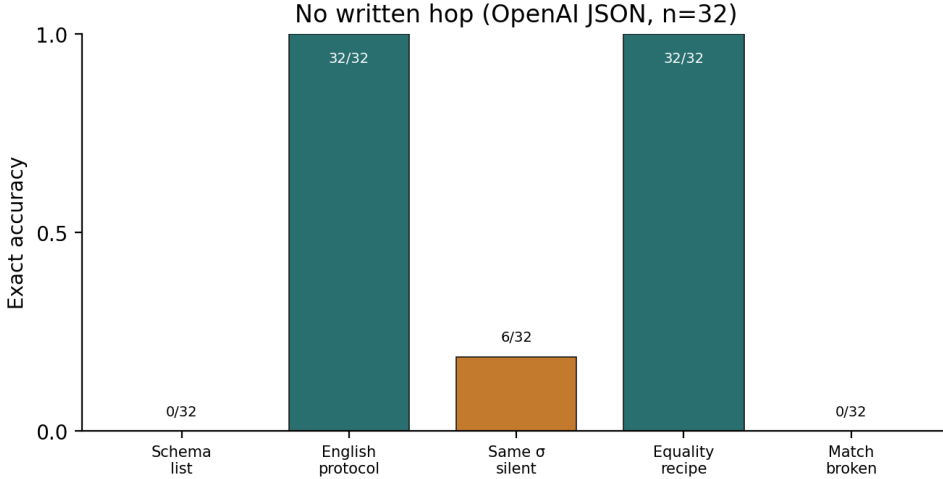


Figure 3: No-hop binding (Wikidata JSON, OpenAI $n=32$).

Figure 3 is the OpenAI row of Table 11. The Table 11 prompts are fixed English blocks (same wording for all three families). We use *equality* rather than *similarity*: exact identity of sealed tokens, not a continuous metric. Schema list only is a bilingual KEY_SCHEMA (English name = sealed JSON key) plus “Use the schema to read CONTEXT. No JOIN is given.” The English protocol adds that QUESTION and CONTEXT are encoded, that identical tokens are the same entity or field (equality, not decryption), and that schema names plus token equality walk CONTEXT; it also names the asked field as country rather than city. The same- σ equality recipe (Appendix K) does not name

country or city in English. It is not the WIKI-H5 prompt. The suite uses the same 32 Wikidata people (per-item keys; seed 20260826–20260829). The OpenAI $n=6$ cell with an English question template and English JSON keys (1/6) is not plaintext Wiki-H5: the start token and all values are sealed, city is a nested distractor under hq, and the prompt forbids world knowledge and allows UNKNOWN. English keys alone do not restore the walk.

F Sealing middleware

The necessary step is binding a typed full name to one graph atom: spaced tokens otherwise miss V . The layer then seals entities, relations, and sensitive values. When the relation sequence is already specified, the join can be executed by a deterministic non-LLM router. If the hop is not compiled, the layer may attach one sealed join as an execution control (not a list of candidate paths). The model, when used, returns $\sigma(a)$; the layer unseals. Isolation on the CEO vault without start-entity binding is 0/6 vs. full layer 6/6 for Composer 2.5, OpenAI, and an unversioned Claude endpoint (Anthropic model id not stored; plaintext names in the packed prompt: 0). On complex 3-hop and real 2Wiki, Composer 2.5 without binding is 0/6 vs. layer 6/6; OpenAI layer 6/6 (without binding: complex 4/6, 2Wiki 0/6). Those isolation files also attach a join: that is the execution control, not the reason to call a model. The model never receives a retrieved path list and never answers in plaintext (Table 1). The 6/6 scores are this binding-and-join control, not the free WIKI-H5 6/32 cell.

G Packaging probes

These cells test serialization and application packaging, not the matched 2×2 . Sample sizes are $n=6-16$ except a counterfactual 2Wiki sanity check and the public Spider identifier-in- q cell ($n=100$).

Tables / JSON / SQL. Same CEO join as markdown tables, nested JSON, or a value store: free sealed English 0/12, join plan 12/12 (Composer 2.5 and OpenAI both 12/12). JSON keys without a legend: 0/6; a key legend yields 6/6 for OpenAI only (6/6 vs. 0/6); a join plan yields 6/6 for both. Token overlap vs. SQLCOL: break on public Spider is Table 12 (OpenAI; identifier-in- q slice $n=100$: 51/100 vs. 0/100). Specified schema→model SQL→local execution on the CEO sqlite skin is 6/6, a different cell.

Excel. The NL arm seals a template that already names both join predicates and the person (batched $n=16$). Composer exact-match 0/16 (mid-hop company-token error 16/16) vs. plan 16/16; OpenAI 16/16 on both. Softer than CEO free sealed English.

FinQA / WTQ (Chen et al., 2021; Pasupat and Liang, 2015). Shared sealed context from 414 tables: FinQA NL 0/8 vs. gold programs 8/8; WTQ NL 0/6 vs. LOOKUP DSL 6/6 (Composer). OpenAI matches FinQA and WTQ DSL; WTQ NL is 1/6. Some programs embed literals or gold cell addresses. Not a leaderboard.

Counterfactual 2Wiki sanity check. Composer 2.5 plain NL 200/200 (Wikipedia leak 0); OpenAI batched 150/200 with overlap leak 50/200. We do not treat this as a standard 2Wiki evaluation. Spider: sealed-cell NL 4/12; model SQL executed locally 29/32; gold-SQL engine 191/200 (Appendix I).

Long context / rename. At $\sim 128k$ and $\sim 1.14M$ sealed tokens, free sealed NL remains 0/8; plans 8/8. Paired σ/σ' sealed plans: 12/12 with differing golds (Figure 4).

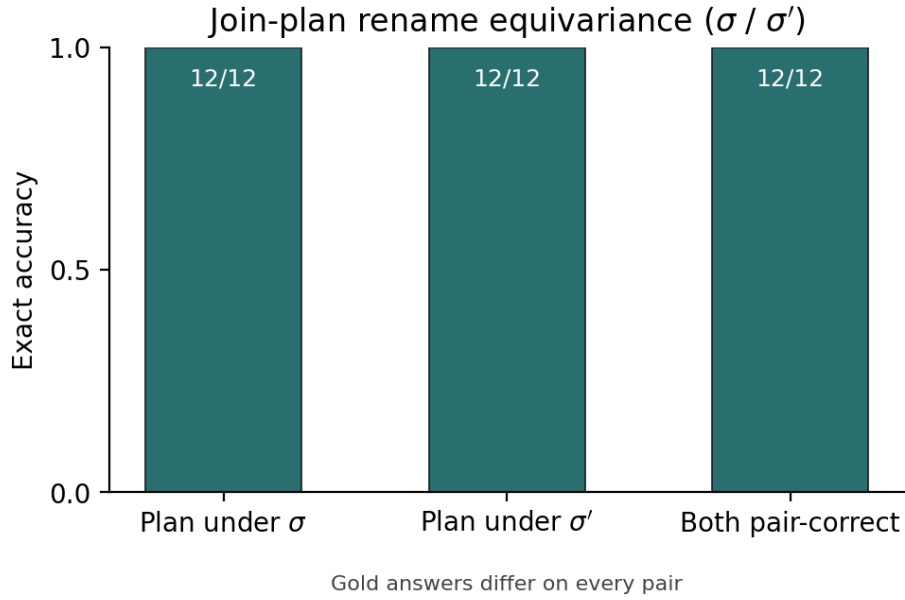


Figure 4: Sealed-plan rename equivariance ($n=12$): plans remain correct under a second key σ' with a different gold on every pair.

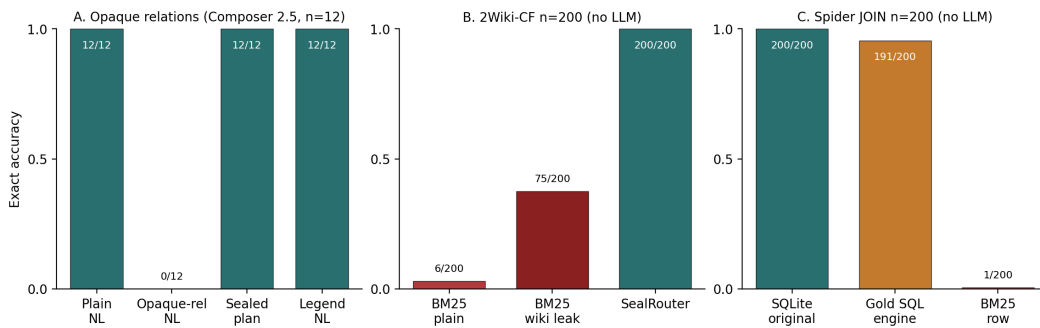


Figure 5: Non-LLM and packaging controls (file fig7_opaque_rel_n200.png). Panel A is Composer $n=12$; panels B–C are frozen $n=200$ baselines. Left: opaque-rel NL 0/12 vs. plan/legend 12/12 (Composer 2.5, $n=12$). Middle: 2Wiki-CF BM25 6/200, Wikipedia-string leak 75/200, SealRouter 200/200. Right: Spider sqlite 200/200, gold-SQL engine 191/200, BM25 1/200. CEO/2Wiki opacity $\times$ prompt-form scores are in the surrounding prose.

English q	Find the number of papers published by the institution “University of Pennsylvania”.
Sealed q (A)	Find the number of E41f626311a9f published by the institution “University of Pennsylvania”.
EQ	q token equals sqlite identifier $\rightarrow$ execute; first-cell gold on this item
NS	same q ; identifiers are HMAC (SQLCOL:name) $\rightarrow$ UNKNOWN
A (name in q)	$n=100$ (45 DBs). EQ 51/100; NS 0/100 (UNKNOWN 100/100)
B (no name in q)	$n=26$. EQ 18/26; NS 20/26 (English q ; legend can still bind)

Table 12: Spider identifier overlap vs. namespace break (OpenAI; Yu et al., 2018). Slice A $n=100$: EQ 51/100, NS 0/100. Not official exact-match. n.r. for Composer/Grok.

H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except [start]: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That $n=32$ template cell is not MOVIE-B.

Secondary cells (not WIKI-H5). n.r. means the cell was not run.

Cell	Prot.	Composer 2.5	OpenAI	Grok 4.5
CEO / 2Wiki plain NL	$n=12$	12/12	12/12	n.r.
2Wiki sealed path	$n=12$	12/12	12/12	n.r.
Vault, no legend	iso. $n=6$	0/6	0/6	n.r.
Vault + legend	iso. $n=6$	6/6	6/6	n.r.
JSON keys: no legend	iso. $n=6$	0/6	0/6	n.r.
JSON keys: + key legend	iso. $n=6$	0/6	6/6	n.r.
JSON keys: + join plan	iso. $n=6$	6/6	6/6	n.r.
Schema list only	iso. $n=32$	0/32	0/32	0/32
English protocol	iso. $n=32$	32/32	32/32	32/32
Same- σ JSON, silent	iso. $n=32$	19/32	6/32	31/32
Same- σ JSON + equality	iso. $n=32$	32/32	32/32	32/32
JSONKEY: no iso match	iso. $n=32$	0/32	0/32	0/32
Sealed-Q two-route, recipe	iso. $n=32$	25/32	31/32	32/32
Sealed-Q two-route, vague	iso. $n=32$	23/32	13/32	31/32
SQLite schema $\rightarrow$ exec	iso. $n=6$	6/6	6/6	n.r.
SQL same- σ overlap (Spider A)	public $n=100$	n.r.	51/100	n.r.
SQL SQLCOL: break (A)	public $n=100$	n.r.	0/100	n.r.
HMAC cols (tables Eng.)	iso. $n=6$	0/6	4/6	n.r.
HMAC tables+cols, no bind	iso. $n=6$	0/6	0/6	n.r.
HMAC tables+cols, bind	iso. $n=6$	6/6	6/6	n.r.
EXCEPT schema exec	iso. $n=6$	6/6	6/6	n.r.
Schema-visible mild NL	$n=12$	12/12	12/12	n.r.
Middleware, full	iso. $n=6$	6/6	6/6	n.r.
Middleware, no start bind	iso. $n=6$	0/6	4/6, 0/6	n.r.
Complex 3-hop sealed NL	iso. $n=8$	8/8	8/8	n.r.
2-hop demos $\rightarrow$ 3-hop	iso. $n=6$	0 $\rightarrow$ 6/6	0 $\rightarrow$ 6/6	n.r.
2-hop demos $\rightarrow$ 4-hop	iso. $n=6$	1 $\rightarrow$ 6/6	1 $\rightarrow$ 6/6	n.r.
Messy-doc extractive	iso. $n=6$	6/6	6/6	n.r.
Opaque-rel iso floor	iso. $n=12$	0/12	0/12	n.r.
2Wiki-CF PLAIN	scale	200/200	150/200	n.r.
Excel NL	batch $n=16$	0/16	16/16	n.r.
Excel plan	batch $n=16$	16/16	16/16	n.r.
FinQA NL	batch $n=8$	0/8	0/8	n.r.
FinQA gold program	batch $n=8$	8/8	8/8	n.r.
FinQA follow	batch $n=8$	4/4	4/4	n.r.
WTQ NL	batch $n=6$	0/6	1/6	n.r.
WTQ DSL	batch $n=6$	6/6	6/6	n.r.
Three-arm plan vs. NL	iso. $n=32$	32/32 vs. 0/32	32/32 vs. 0/32	32/32 vs. 0/32
Dual-path matched vs. novel	iso. $n=32$	note	note	note

Table 13: Secondary-cell ledger. 0 $\rightarrow$ 6/6 = zero-demo then demos (6/6); 4-hop zero-demo is 1/6. n.r. = not run. JSON key legend: OpenAI 6/6, Composer 0/6; join plan 6/6 for both. HMAC cols (tables English) is a table-name leak (OpenAI 4/6). Middleware without start binding: OpenAI complex 4/6, 2Wiki 0/6 (full middleware 6/6). 2Wiki-CF PLAIN: batched leak 50/200; isolation leak shard OpenAI 48/50 leak 2/50. Excel NL seals Where is the company headquartered_in that {person} works_at?; after sealing, start and both relation atoms appear in the question (batched $n=16$). The template already names the start and both join predicates, so the arm is softer than CEO free sealed English 0/12. Dual-path free-form: Composer matched 32/32, novel bal. UNK 32/32, novel asym. decoy 32/32, plan 32/32; OpenAI matched 32/32, novel bal. UNK 32/32, novel asym. 0/32 (UNK 24, decoy 8), plan 32/32; Grok matched 32/32, novel UNK 32/32 (both noise regimes), plan 32/32. Spider A (Table 12): public identifier-in- q slice; first-cell denotation; OpenAI only. Slice B is not this contrast.

I Symbolic execution controls

SealRouter executes a gold sealed path with no LLM. The gold-SQL engine executes gold Spider SQL locally. Figure 5 summarizes those controls.

J Sign-flip of the H5 interaction

Let $d_i = \mathbf{1}[\text{opaque correct}] - \mathbf{1}[\text{English correct}]$ and $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$. Of 32 WIKI-H5 items, 24 have $c_i = -1$, one has $c_i = -2$, 7 have $c_i = 0$; $T = -26$. Under exchangeability of unique-path and two-path opacity effects, $p = 2/2^{25} = 5.96 \times 10^{-8}$. That p coincides numerically with the two-path McNemar p (25 vs. 0); the tests are not the same. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

K Equality recipe (JSON cell)

Verbatim protocol used in Table 11, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”